

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1186

Signetics 100 kV
July 01, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1186. The Interconnection Customer proposes a 181.7 MW solar plus storage facility interconnecting at the Signetics 100 kV substation in the Indiana area. The request was received on April 10, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 426 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,294,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Signetics - Montpelier Electric Generating Sta 100 KV line	Hartford City - Signetics 100 KV Line	171.1	142	92.5
Hartford City - Signetics 100 KV	Signetics - Montpelier Electric Generating Station 100 KV L	169.3	126	80.1
Intalco - Montpelier Electric Generating Station Substation 100 KV	Signetics - Montpelier Electric Generating Sta 100 KV line	107.4	130	59.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Signetics - Montpelier Electric Generating Sta 100 KV line	Rebuild line with higher-capacity conductor	\$2,641,000	58% (\$1,524,000)
Hartford City - Signetics 100 KV	Rebuild line with higher-capacity conductor	\$12,902,000	58% (\$7,494,000)
Intalco - Montpelier Electric Generating Station Substation 100 KV	Reconductor line	\$2,375,000	54% (\$1,276,000)
	Total allocated to project		\$10,294,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.